

Examiner
only

- 9.** (a) The following diagram shows an outline of part of the Periodic Table.

The letters shown are NOT the chemical symbols of the elements.

[illegible]

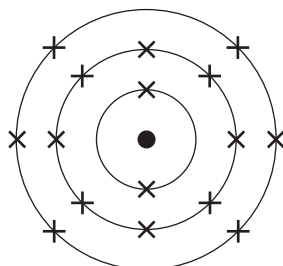
Choose **letters** from the diagram to complete the table below.

[4]

	Letter
The element in Group 3 and Period 2	
The element which has 10 protons in its nucleus	
The element with the electronic structure 2,8,6	
The element which forms a 2+ ion	



- (b) The diagram below shows the electronic structure of an element in the Periodic Table.



In the space below, draw a diagram to show the electronic structure of the element which lies directly **above** it.

[1]

- (c) The table shows information about atoms **X**, **Y** and **Z**.

Atom	Symbol	Number of protons	Number of neutrons	Number of electrons
X	$^{31}_{15}\text{X}$		16	15
Y	$^{39}_{19}\text{Y}$	19		19
Z	$^{40}_{19}\text{Z}$	19	21	

- (i) Complete the table.

[3]

- (ii) Underline the term used to describe atoms **Y** and **Z**.

[1]

ions

inert

insoluble

isotopes

